

Managing Climate Risk: Sovereign and Corporate Debt Markets

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Abstract

Physical climate risks such as droughts, floods, and extreme heat are increasingly recognized as systematic risk factors for the real economy and financial markets, yet we know comparatively little about whether and how public policy can reshape their pricing.

Keywords: Physical climate risks, asset pricing, public policy

1. Introduction

2. Model

2.1. Economic Environment, Climate Risk, and Adaptation

Time is discrete, $t = 0, 1, 2, \dots$. Uncertainty is generated by a climate risk factor X_t and other aggregate shocks. Government adaptation investment is denoted by

$G_t \geq 0$, which can be interpreted as the flow of public spending on climate adaptation (e.g., large-scale water projects, flood defenses). For now, we treat G_t as exogenous and focus on its asset-pricing implications.

Climate risk dynamics. We model the climate factor X_t as a (possibly persistent) shock that captures the intensity of physical climate risk (e.g., drought risk) at time t . Its dynamics are given by

$$X_{t+1} = (1 - \phi(G_t))X_t + \sigma(G_t)\varepsilon_{t+1}, \quad (1)$$

where ε_{t+1} is an i.i.d. innovation with mean zero and unit variance, $\phi(G_t) \in [0, 1]$ measures the mean-reversion of the climate shock, and $\sigma(G_t) > 0$ is the conditional volatility of X_{t+1} .

We assume:

$$\phi'(G) > 0, \quad \sigma'(G) < 0. \quad (2)$$

That is, a higher level of adaptation investment G increases the mean-reversion of climate shocks and reduces their conditional variance. Under (1)–(2), G reduces both the persistence and volatility of climate risk.

Aggregate output and tax base. Aggregate output next period is given by

$$Y_{t+1} = \bar{Y} - d_X X_{t+1} + u_{t+1}, \quad (3)$$

where $d_X > 0$ captures the adverse impact of climate risk on output and u_{t+1} is an i.i.d. disturbance. The government collects proportional taxes

$$T_{t+1} = \tau Y_{t+1}, \quad \tau \in (0, 1). \quad (4)$$

Stochastic discount factor. Asset prices are determined by a stochastic discount factor (SDF) M_{t+1} . We do not commit to a particular preference specification but assume that climate risk is *bad* in the sense that

$$\text{Cov}_t(M_{t+1}, X_{t+1}) > 0. \quad (5)$$

Intuitively, high realizations of X_{t+1} (more severe climate risk) are associated with high marginal utility and thus high M_{t+1} .

2.2. Government Sector and Sovereign Debt Pricing

The government issues one-period defaultable bonds. Let B_t denote the stock of outstanding government debt at time t , and let r_t^f be the risk-free short rate. The government budget constraint is

$$B_{t+1} = (1 + r_t^g)B_t + G_t + I_t - T_t, \quad (6)$$

where r_t^g is the required return on sovereign bonds, G_t is adaptation spending, and I_t collects other government expenditures.

Default intensity. We model sovereign default in reduced form. Let $D_{t+1}^g \in \{0, 1\}$ be the indicator that the government defaults between t and $t + 1$. The (physical) default intensity is

$$\lambda_t^g = \Lambda^g(X_t, B_t, G_t), \quad (7)$$

where $\Lambda^g(\cdot)$ is increasing in climate risk and debt, but decreasing in adaptation:

$$\frac{\partial \Lambda^g}{\partial X} > 0, \quad \frac{\partial \Lambda^g}{\partial B} > 0, \quad \frac{\partial \Lambda^g}{\partial G} < 0. \quad (8)$$

These assumptions capture the idea that severe climate risk and high debt raise sovereign default risk, while greater adaptation investment improves macroeconomic resilience and reduces default intensity.

Sovereign bond pricing and spreads. Consider a one-period sovereign bond with face value one and price P_t^g . Its payoff is $(1 - D_{t+1}^g)$ in units of the numeraire. The no-arbitrage pricing equation is

$$P_t^g = \mathbb{E}_t \left[M_{t+1} (1 - D_{t+1}^g) \right]. \quad (9)$$

Define the gross required return

$$1 + r_t^g := \frac{1}{P_t^g}. \quad (10)$$

Let $P_t^f = \mathbb{E}_t[M_{t+1}]$ denote the price of a default-free one-period bond; its return satisfies $1 + r_t^f = 1/P_t^f$. The *sovereign spread* is defined as

$$s_t^g := r_t^g - r_t^f. \quad (11)$$

Under a standard linearization of (9), the sovereign spread can be decomposed into an expected-loss component and a risk-premium component linked to climate risk.¹

Proposition 1 (Decomposition of sovereign spreads) *Under mild regularity conditions and a local linearization around the mean, the sovereign spread can be written as*

$$s_t^g \approx \underbrace{\lambda_t^{g,\mathbb{P}}}_{\text{expected loss}} + \underbrace{\beta_X^g(G_t) \lambda_X(G_t)}_{\text{climate risk premium}} + (\text{other risk premia}), \quad (12)$$

where $\lambda_t^{g,\mathbb{P}}$ is the physical default intensity, $\beta_X^g(G_t)$ is the sovereign bond's exposure (beta) to the climate factor X_{t+1} , and $\lambda_X(G_t)$ is the market price of climate risk. Both $\beta_X^g(G)$ and $\lambda_X(G)$ are functions of the level of adaptation G through the climate dynamics (1)–(2) and the SDF covariance (5).

We next characterize how an increase in adaptation G_t affects the sovereign spread.

Proposition 2 (Climate adaptation and sovereign financing costs) *Suppose that:*

¹The exact functional form is not central for our comparative statics. A formal derivation is provided in the Appendix XX.

1. Λ^g satisfies the monotonicity conditions (8);
2. G affects the climate dynamics through (1)–(2);
3. the SDF satisfies (5).

Then, holding (X_t, B_t) fixed, the sovereign spread s_t^g is decreasing in G_t :

$$\frac{\partial s_t^g}{\partial G_t} < 0. \quad (13)$$

The total effect can be decomposed into:

$$\frac{\partial s_t^g}{\partial G_t} = \underbrace{\frac{\partial \lambda_t^{g, \mathbb{P}}}{\partial G_t}}_{<0, \text{ expected loss effect}} + \underbrace{\frac{\partial \beta_X^g(G_t)}{\partial G_t} \lambda_X(G_t) + \beta_X^g(G_t) \frac{\partial \lambda_X(G_t)}{\partial G_t}}_{\leq 0, \text{ risk premium effect}}.$$

Proof sketch. The expected-loss effect follows directly from (7)–(8). The risk-premium effect arises because G reduces both the variance of X_{t+1} and its persistence, thereby lowering the sovereign bond’s exposure $\beta_X^g(G)$, and because the market price of climate risk $\lambda_X(G)$ is decreasing in the volatility of X_{t+1} given (5). A formal derivation is provided in Appendix XX. ■

Corollary 1 (Cross-sectional heterogeneity in sovereign spread relief) *The impact of adaptation investment G_t on sovereign spreads is stronger when:*

1. the economy is more climate-exposed (higher d_X in (3)), and/or
2. the government debt level B_t is higher (so that a given reduction in default intensity generates a larger change in expected loss).

2.3. Corporate Sector and Credit Spreads

We now introduce a continuum of firms indexed by i . Each firm issues defaultable corporate bonds and is exposed to both climate risk and sovereign risk.

Corporate cash flows and climate exposure. Firm i 's cash flow next period is

$$CF_{i,t+1} = \bar{C}F_i - \theta_i X_{t+1} - \eta_i \lambda_t^g + \epsilon_{i,t+1}, \quad (14)$$

where $\theta_i \geq 0$ measures the firm's technological exposure to climate risk (e.g., water-intensive production), $\eta_i \geq 0$ captures its exposure to sovereign risk (e.g., dependence on fiscal transfers or regulated tariffs), and $\epsilon_{i,t+1}$ is an idiosyncratic shock.

Corporate default intensity. Firm i 's (physical) default intensity is modeled as

$$\lambda_{i,t}^c = \Lambda^c(X_t, G_t; Z_i), \quad (15)$$

where Z_i collects firm-level characteristics. We assume

$$\frac{\partial \Lambda^c}{\partial X} > 0, \quad \frac{\partial \Lambda^c}{\partial G} < 0, \quad (16)$$

so that worse climate conditions raise default risk, while adaptation investment attenuates it.

Corporate bond pricing and spreads. Consider a one-period zero-coupon bond issued by firm i with price $P_{i,t}^c$ and payoff $(1 - D_{i,t+1}^c)$. The price satisfies

$$P_{i,t}^c = \mathbb{E}_t \left[M_{t+1} (1 - D_{i,t+1}^c) \right], \quad (17)$$

where $D_{i,t+1}^c$ is the default indicator. The corporate bond return is $1 + r_{i,t}^c := 1/P_{i,t}^c$, and the corporate credit spread is

$$s_{i,t}^c := r_{i,t}^c - r_t^f. \quad (18)$$

Parallel to Proposition 1 and Proposition 2, we obtain:

Proposition 3 (Decomposition of corporate credit spreads) *Under a local linearization, the corporate credit spread can be written as*

$$s_{i,t}^c \approx \underbrace{\lambda_{i,t}^{c,\mathbb{P}}}_{\text{expected loss}} + \underbrace{\beta_{i,X}^c(G_t) \lambda_X(G_t)}_{\text{climate risk premium}} + (\text{other risk premia}), \quad (19)$$

where $\lambda_{i,t}^{c,\mathbb{P}}$ is the firm's physical default intensity, $\beta_{i,X}^c(G_t)$ is the firm's bond exposure to climate risk, and $\lambda_X(G_t)$ is the same market price of climate risk as in (12).

Proposition 4 (Climate adaptation and corporate financing costs) *If (16) holds and G affects the climate dynamics as in (1)–(2), then, holding (X_t, Z_i) fixed, the corporate credit spread $s_{i,t}^c$ is decreasing in adaptation investment:*

$$\frac{\partial s_{i,t}^c}{\partial G_t} < 0. \quad (20)$$

The reduction in spreads is stronger for firms with larger climate-exposure coefficients θ_i and sovereign-risk loadings η_i in (14), as summarized in Corollary 2.

Corollary 2 (Cross-sectional predictions for corporate spreads) *For a given change in adaptation investment G_t , the decline in credit spreads $s_{i,t}^c$ is predicted to be:*

1. *larger for firms with higher θ_i (more climate-sensitive cash flows, e.g., water-intensive industries); and*
2. *larger for firms with higher η_i (stronger dependence on sovereign risk, e.g., state-owned enterprises or regulated utilities).*

These cross-sectional predictions provide testable implications for the empirical analysis.

2.4. Joint Government–Corporate Financing Conditions

Finally, define the average sovereign spread $\bar{s}_t^g := \mathbb{E}[s_t^g]$ and an aggregate corporate spread $\bar{s}_t^c := \mathbb{E}_i[s_{i,t}^c]$. Combining Propositions 2 and 4, we obtain:

Proposition 5 (Adaptation investment and aggregate financing costs) *Under the above assumptions, an increase in adaptation investment G_t reduces both sovereign and corporate spreads at the aggregate level:*

$$\frac{\partial \bar{s}_t^g}{\partial G_t} < 0, \quad \frac{\partial \bar{s}_t^c}{\partial G_t} < 0. \quad (21)$$

Moreover, the joint reduction in public and private financing costs can offset part of the contemporaneous fiscal cost of adaptation G_t in the government budget constraint (6).

2.5. Government's Optimal Adaptation Policy

In the previous subsections, government adaptation investment G_t is taken as exogenous. We now endogenize G_t by specifying a multi-objective government problem in which the policy maker trades off three goals: (i) sustaining aggregate economic activity, (ii) mitigating climate damages, and (iii) improving sovereign and corporate financing conditions.

Per-period social objective. Let C_t denote aggregate (private) consumption and let $D(X_t)$ capture the flow disutility from climate risk X_t , with $D'(X) > 0$ and $D''(X) \geq 0$.² We also let high sovereign and corporate spreads generate additional welfare costs, as they tighten public and private financing constraints. The per-period social welfare is

$$W_t = U(C_t) - \alpha D(X_t) - \gamma^g s_t^g B_t - \gamma^c \bar{s}_t^c K_t, \quad (22)$$

where, $U(\cdot)$ is a standard increasing, concave utility function, $U'(C) > 0$, $U''(C) < 0$; $\alpha > 0$ measures the weight on reducing climate damages; $\gamma^g > 0$ and $\gamma^c > 0$ measure the social costs of high sovereign and corporate financing spreads, respectively; s_t^g is

²Modeling climate damages partly as utility/amenity damages is standard in the climate-economy literature; see, e.g., Barrage (2020).

the sovereign spread defined in (11), B_t is government debt; $\bar{s}_t^c$ is an aggregate corporate spread defined in (19); K_t measures the aggregate scale of private borrowing or capital.

The multi-objective nature of the problem is made explicit in (22): the government puts (implicit) weights on consumption, climate damages, and financing conditions. In practice, this is equivalent to a standard single-objective optimization with a weighted composite criterion.

Resource constraint and financing. For concreteness, we assume that aggregate consumption is given by

$$C_t = Y_t - G_t - \Phi(B_t, s_t^g) - \Psi(K_t, \bar{s}_t^c), \quad (23)$$

where Y_t is aggregate output defined in (3), G_t is adaptation spending, and $\Phi(\cdot)$ and $\Psi(\cdot)$ capture government and private resources absorbed by debt service and financing frictions, with

$$\frac{\partial \Phi}{\partial s^g} > 0, \quad \frac{\partial \Psi}{\partial \bar{s}^c} > 0.$$

Thus, higher sovereign and corporate spreads tighten the resource constraint by increasing required payments and reducing available consumption.

Government debt evolves according to the budget constraint (6), repeated here for convenience:

$$B_{t+1} = (1 + r_t^g)B_t + G_t + I_t - T_t, \quad (24)$$

with $r_t^g = r_t^f + s_t^g$.

Dynamic programming problem. The government chooses a sequence $\{G_t\}_{t \geq 0}$ to maximize the expected discounted sum of social welfare:

$$\max_{\{G_t\}_{t \geq 0}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t W_t, \quad \beta \in (0, 1), \quad (25)$$

subject to the climate dynamics (1), the output equation (3), the government budget constraint (24), the bond pricing relations (9) and (17), and the definitions of sovereign and corporate spreads in (11) and (19).

Let the state vector be $S_t = (X_t, B_t, \Xi_t)$, where Ξ_t collects other relevant aggregate states (e.g., K_t and exogenous macro shocks). The government's value function satisfies

$$V(S_t) = \max_{G_t \geq 0} \left\{ W_t(S_t, G_t) + \beta \mathbb{E}_t[V(S_{t+1})] \right\}, \quad (26)$$

where S_{t+1} evolves according to the laws of motion implied by the constraints and the climate dynamics (1).

First-order condition for optimal adaptation. Assuming an interior solution and differentiability, the first-order condition (FOC) for G_t can be written as

$$\frac{\partial W_t}{\partial G_t} + \beta \mathbb{E}_t \left[V_X(S_{t+1}) \frac{\partial X_{t+1}}{\partial G_t} + V_B(S_{t+1}) \frac{\partial B_{t+1}}{\partial G_t} + V_{\Xi}(S_{t+1}) \frac{\partial \Xi_{t+1}}{\partial G_t} \right] = 0. \quad (27)$$

Using (22), (23), and (24), the current-period derivative $\partial W_t/\partial G_t$ can be decomposed as

$$\begin{aligned}\frac{\partial W_t}{\partial G_t} &= U_C(C_t) \frac{\partial C_t}{\partial G_t} - \alpha D'(X_t) \frac{\partial X_t}{\partial G_t} - \gamma^g \left(\frac{\partial s_t^g}{\partial G_t} B_t + s_t^g \frac{\partial B_t}{\partial G_t} \right) - \gamma^c \left(\frac{\partial \bar{s}_t^c}{\partial G_t} K_t + \bar{s}_t^c \frac{\partial K_t}{\partial G_t} \right) \\ &= \underbrace{U_C(C_t) \frac{\partial C_t}{\partial G_t}}_{\text{utility-cost}} + \underbrace{\gamma^g B_t \left(-\frac{\partial s_t^g}{\partial G_t} \right) + \gamma^c K_t \left(-\frac{\partial \bar{s}_t^c}{\partial G_t} \right)}_{\text{current financing-cost relief}}.\end{aligned}\quad (28)$$

At time t , (X_t, B_t, K_t, Ξ_t) are predetermined state variables. Hence, $\partial X_t/\partial G_t = \partial B_t/\partial G_t = \partial K_t/\partial G_t = 0$. Policy G_t affects welfare contemporaneously through resource allocation ($\partial C_t/\partial G_t$), current financing-cost relief and also through the continuation value via next-period states, notably $\partial X_{t+1}/\partial G_t$, $\partial B_{t+1}/\partial G_t$, and $\partial \Xi_{t+1}/\partial G_t$ in (27). We want to emphasize that although the current climate damage reduction terms is zero, i.e., $-\alpha D'(X_t) \frac{\partial X_t}{\partial G_t} = 0$, but there are similar future benefit via continuation value term through $\partial X_{t+1}/\partial G_t$.

Note that,

$$\underbrace{\frac{\partial C_t}{\partial G_t}}_{\text{net resource-cost}} = \underbrace{-1}_{\text{direct resource-cost} < 0} + \underbrace{\left[\Phi_{s^g} \left(-\frac{\partial s_t^g}{\partial G_t} \right) + \Psi_{\bar{s}^c} \left(-\frac{\partial \bar{s}_t^c}{\partial G_t} \right) \right]}_{\text{resource-release via spread compression} > 0}.$$

Policy G_t can reduce the resource cost through resource release via spread compression, hence effectively reducing the net resource-cost. Hence, (28) can be rewritten

as follows

$$\frac{\partial W_t}{\partial G_t} = U_C(C_t) \underbrace{\left[-1 + \Phi_{sg} \left(-\frac{\partial s_t^g}{\partial G_t} \right) + \Psi_{\bar{s}^c} \left(-\frac{\partial \bar{s}_t^c}{\partial G_t} \right) \right]}_{\text{net utility cost}} + \underbrace{\gamma^g B_t \left(-\frac{\partial s_t^g}{\partial G_t} \right) + \gamma^c K_t \left(-\frac{\partial \bar{s}_t^c}{\partial G_t} \right)}_{\text{current financing-cost relief } > 0}. \quad (29)$$

Combining (27) and (29), the optimality condition equates the marginal resource cost of adaptation to its marginal benefits in terms of improved public–private financing conditions, plus discounted future benefits, e.g., future climate damages relief. We summarize this characterization as follows.

Proposition 6 (Characterization of optimal adaptation) *Suppose Assumptions (2), (8), and (16) hold, so that adaptation G_t reduces the persistence and volatility of climate risk and decreases both sovereign and corporate default intensities. Then the government’s optimal adaptation policy G_t^* satisfies*

$$U_C(C_t) \underbrace{\left[1 - \Phi_{sg} \left(-\frac{\partial s_t^g}{\partial G_t} \right) - \Psi_{\bar{s}^c} \left(-\frac{\partial \bar{s}_t^c}{\partial G_t} \right) \right]}_{\text{net resource-cost}} = \underbrace{\gamma^g B_t \left(-\frac{\partial s_t^g}{\partial G_t} \right) + \gamma^c K_t \left(-\frac{\partial \bar{s}_t^c}{\partial G_t} \right)}_{\text{current benefit: financing-cost relief}} + \underbrace{\beta \mathbb{E}_t \left[V_X(S_{t+1}) \frac{\partial X_{t+1}}{\partial G_t} + V_B(S_{t+1}) \frac{\partial B_{t+1}}{\partial G_t} + V_{\Xi}(S_{t+1}) \frac{\partial \Xi_{t+1}}{\partial G_t} \right]}_{\text{intertemporal benefit term}} \quad (30)$$

In particular, the net marginal utility cost of raising adaptation by one unit (left-hand side) is equal to the sum of two terms: (i) *current* benefits from improved financing conditions—lower sovereign and corporate spreads weighted by the scale of public and private borrowing—and (ii) an *intertemporal* benefit term that captures

how G_t shifts next-period states and thereby the continuation value.

The intertemporal term comprises multiple channels. First, adaptation reduces future climate risk exposure through $\partial X_{t+1}/\partial G_t$, lowering future climate-related welfare losses embedded in $V_X(S_{t+1})$. Second, adaptation affects debt dynamics through $\partial B_{t+1}/\partial G_t$ by compressing sovereign spreads and improving rollover conditions. Higher G_t can improve future debt sustainability and thus future fiscal space/market access (captured by $V_B(S_{t+1})$). More generally, $V_\Xi(S_{t+1})\partial\Xi_{t+1}/\partial G_t$ captures additional discounted gains from changes in other aggregate states (e.g., private-sector financing capacity or fiscal capacity shifters) induced by adaptation.

Corollary 3 (Comparative statics for the optimal adaptation level) *Consider a stationary environment in which the state vector S_t is constant and the optimal adaptation level is G^* . Then:*

1. *If the economy becomes more climate-exposed (higher d_X in (3) or a steeper damage function $D(\cdot)$), the optimal adaptation level G^* increases.*
2. *If the social weights on financing costs (γ^g, γ^c) increase, or if the scale of public and private borrowing (B_t, K_t) increases, the optimal adaptation level G^* increases.*
3. *If adaptation becomes more effective in reducing spreads (i.e., $\partial s_t^g/\partial G_t$ and $\partial \bar{s}_t^c/\partial G_t$ become more negative), the optimal adaptation level G^* increases.*

These comparative statics highlight that stronger climate exposure and higher debt burdens push the government towards more aggressive adaptation investment from the perspective of joint public–private financing conditions.

2.6. Fiscal Capacity Constraint and Loop Identification

2.6.1. An explicit debt limit and the shadow price of fiscal space

To make “fiscal space” operational in the model, we impose an explicit fiscal-capacity (market-access) constraint. Specifically, in addition to the government budget dynamics (24), we assume a debt limit (or an equivalent market-access constraint):

$$B_{t+1} \leq \bar{B}(\Xi_t), \quad (31)$$

where $\bar{B}(\Xi_t)$ captures time-varying fiscal capacity (institutional quality, external liquidity, rollover needs) summarized by Ξ_t .³ Let $\mu_t \geq 0$ be the associated Lagrange multiplier. We interpret μ_t as the *shadow price of fiscal space*: $\mu_t = 0$ when unconstrained, and $\mu_t > 0$ when fiscal capacity binds.

2.6.2. Optimal adaptation with a fiscal-space wedge

Under the debt dynamics (24), the marginal effect of G_t on next-period debt is

$$\frac{\partial B_{t+1}}{\partial G_t} = 1 + B_t \frac{\partial r_t^g}{\partial G_t} = 1 + B_t \frac{\partial s_t^g}{\partial G_t}, \quad (32)$$

where $\partial s_t^g / \partial G_t < 0$ by Proposition 2. The debt-limit multiplier μ_t therefore enters the optimal condition for G_t as an additional wedge: relative to (30) in Proposition 6,

³All results below have a one-to-one analogue under a market-access constraint such as $s_t^g \leq \bar{s}(\Xi_t)$ or $r_t^g \leq \bar{r}(\Xi_t)$, with a multiplier attached to the spread/interest-rate cap.

the marginal cost side is shifted by

$$\mu_t \left(1 + B_t \frac{\partial s_t^g}{\partial G_t} \right). \quad (33)$$

In other words, the first-order condition for G_t is modified relative to Proposition 6 by an additional fiscal-space wedge:

$$\underbrace{U_C(C_t) \frac{\partial C_t}{\partial G_t}}_{\text{marginal cost}} + \underbrace{\mu_t \left(1 + B_t s_G^g \right)}_{\text{fiscal-space wedge}} = \underbrace{\gamma^g B_t \left(- \frac{\partial s_t^g}{\partial G_t} \right) + \gamma^c K_t \left(- \frac{\partial \bar{s}_t^c}{\partial G_t} \right)}_{\text{current benefit: financing-cost relief}} + \beta \mathbb{E}_t [\mathcal{MB}_{t+1}^{\text{future}}]. \quad (34)$$

The term $\mu_t(1 + B_t s_G^g)$ captures the marginal *tightening/relaxation* of fiscal capacity caused by an additional unit of adaptation spending, accounting for the endogenous interest-rate/spread response.

When the constraint binds ($\mu_t > 0$), adaptation becomes harder to sustain if increasing G_t mechanically raises B_{t+1} faster than it saves on financing costs; conversely, if spread relief is sufficiently strong, higher G_t can *relax* the constraint and reduce μ_t , creating a virtuous cycle.

2.6.3. Threshold inequalities: doom loops, virtuous loops, and multiplicity

We now provide transparent threshold inequalities that identify parameter regions corresponding to doom loops, virtuous cycles, and potential multiplicity/tipping.

Corollary 4 (Net-resource (“self-financing”) condition) *Using the resource*

constraint (23), the marginal impact of G_t on contemporaneous residual resources is

$$\frac{\partial C_t}{\partial G_t} = -1 + \Phi_{sg}(B_t, s_t^g) \left(-\frac{\partial s_t^g}{\partial G_t} \right) + \Psi_{\bar{s}^c}(K_t, \bar{s}_t^c) \left(-\frac{\partial \bar{s}_t^c}{\partial G_t} \right). \quad (35)$$

Hence, adaptation is weakly self-financing on impact (i.e., $\partial C_t / \partial G_t \geq 0$) whenever

$$\Phi_{sg} \left(-\frac{\partial s_t^g}{\partial G_t} \right) + \Psi_{\bar{s}^c} \left(-\frac{\partial \bar{s}_t^c}{\partial G_t} \right) \geq 1. \quad (36)$$

When (36) holds, contemporaneous fiscal slack is not reduced by a higher G_t , making it less likely that fiscal capacity binds (i.e., less likely that $\mu_t > 0$).

Corollary 5 (Debt-improving (constraint-relaxing) condition) *Under the debt dynamics (24), adaptation relaxes the debt limit (31) (i.e., lowers B_{t+1}) if and only if*

$$\frac{\partial B_{t+1}}{\partial G_t} < 0 \iff 1 + B_t \frac{\partial s_t^g}{\partial G_t} < 0 \iff B_t \left(-\frac{\partial s_t^g}{\partial G_t} \right) > 1. \quad (37)$$

When (37) holds, even if the fiscal-capacity constraint binds, a higher G_t reduces next-period debt through spread relief, pushing μ_t down and supporting a virtuous policy-finance feedback.

When (37) holds, even if $\mu_t > 0$, the fiscal-space wedge in (34) becomes *negative*, turning the fiscal constraint into an additional incentive to invest in G_t .

Corollary 6 (Feedback strength and multiplicity (tipping) condition) *Consider the constrained region where $\mu_t > 0$ and the debt limit binds. Let the local tightness schedule be summarized by $\mu_t = \mathcal{M}(B_{t+1} - \bar{B}(\Xi_t))$, with $\mathcal{M}'(\cdot) > 0$ around the*

boundary, and let the constrained-policy best response be $G_t^* = \mathcal{G}(\mu_t)$. A sufficient condition for local multiplicity/tipping is a strong two-way feedback:

$$|\mathcal{G}'(\mu_t)| \cdot |\mathcal{M}'(\cdot)| \cdot \left| 1 + B_t \frac{\partial s_t^g}{\partial G_t} \right| > 1, \quad (38)$$

in which case a small deterioration in fiscal tightness can trigger a discrete drop in G_t^* (doom-loop regime), while the relaxed region supports a high- G_t^* (virtuous) regime.

Corollaries 4–6 jointly identify three empirically meaningful regions. First, the *doom-loop region* is characterized by weak spread-reduction or small debt outstanding such that $1 + B_t s_G^g > 0$ (i.e., $B_t(-s_G^g) < 1$), implying that a binding constraint ($\mu_t > 0$) raises the effective marginal cost of G_t in (34). In this region, adverse climate shocks that increase spreads (and thus tighten fiscal capacity) reduce G_t^* , which in turn slows mitigation of physical risk and perpetuates high spreads (see the doom-loop synthesis in Zenios, 2024). Second, the *virtuous-loop region* emerges when spread reductions are sufficiently strong, $B_t(-s_G^g) > 1$, so that higher G_t lowers B_{t+1} and relaxes the constraint, pushing μ_t down and reinforcing a high- G_t policy. Finally, when the tightness schedule is steep and the policy response to μ_t is strong, (38) implies potential multiplicity/tipping between these regimes.

Proposition 7 (Doom vs. virtuous loop under a fiscal capacity constraint)

Consider the fiscal-capacity (market-access) constraint

$$B_{t+1} \leq \overline{B}(\Xi_t), \quad (39)$$

with associated multiplier $\mu_t \geq 0$. Let $s_G^g \equiv \partial s_t^g / \partial G_t < 0$ denote the sovereign spread response to adaptation, and recall that under the debt dynamics,

$$\frac{\partial B_{t+1}}{\partial G_t} = 1 + B_t s_G^g. \quad (40)$$

Suppose an interior solution for G_t and differentiability so that the (constrained) first-order condition can be written as the unconstrained optimality condition plus the fiscal-space wedge $\mu_t(1 + B_t s_G^g)$.

Then, the economy admits three locally distinct regimes:

(i) Doom-loop regime (vicious cycle). If the fiscal-capacity constraint binds ($\mu_t > 0$) and spread relief is not strong enough to offset the mechanical debt effect,

$$1 + B_t s_G^g > 0 \quad \Longleftrightarrow \quad B_t(-s_G^g) < 1, \quad (41)$$

then the fiscal-space wedge is strictly positive and increases the effective marginal cost of adaptation. In this region, adverse shocks that tighten fiscal capacity (raise μ_t) reduce the optimal adaptation policy G_t^* , which in turn weakens spread compression and climate-risk mitigation, reinforcing tight fiscal space and high spreads.

(ii) Virtuous-cycle regime (self-reinforcing improvement). If spread relief is sufficiently strong so that adaptation reduces next-period debt,

$$1 + B_t s_G^g < 0 \quad \Longleftrightarrow \quad B_t(-s_G^g) > 1, \quad (42)$$

then higher adaptation relaxes fiscal capacity by lowering B_{t+1} , and the fiscal-space

wedge becomes negative whenever $\mu_t > 0$, turning a binding constraint into an additional incentive to invest in G_t . This generates a self-reinforcing policy–finance feedback consistent with a virtuous cycle.

(iii) Multiplicity/tipping region. Let the constrained best response be summarized by $G_t^* = \mathcal{G}(\mu_t; S_t)$ and let $\mu_t = \mathcal{M}(B_{t+1} - \bar{B}(\Xi_t))$ locally around the boundary with $\mathcal{M}'(\cdot) > 0$. If the local two-way feedback satisfies

$$\left| \partial \mathcal{G}(\mu_t; S_t) / \partial \mu_t \right| \left| \mathcal{M}'(\cdot) \right| \left| 1 + B_t s_G^g \right| > 1, \quad (43)$$

then small changes in fiscal tightness can trigger a discrete shift in optimal adaptation (tipping), and the model can exhibit locally multiple equilibria corresponding to low- G_t^* (doom-loop) and high- G_t^* (virtuous-cycle) regimes.

Finally, Corollary 4 (self-financing on impact) and the corporate-sector channel enlarge the set of states in which the economy operates in the virtuous-cycle region by strengthening spread compression and relaxing resource absorption through $\Psi(K_t, \bar{s}_t^c)$.

2.6.4. Managing insights: steering the economy away from the doom loop

Figure 1 provides a transparent regime map for the policy–finance feedback implied by Proposition 7. The key state-dependent object is $\theta_t \equiv B_t(-\partial s_t^g / \partial G_t)$, which summarizes whether adaptation is sufficiently potent in compressing sovereign spreads to improve debt dynamics. When fiscal capacity does *not* bind ($\mu_t = 0$, Panel A), the main question is whether adaptation is self-financing on impact, captured by $\rho_t \equiv \Phi_{sg}(-\partial s_t^g / \partial G_t) + \Psi_{sc}(-\partial \bar{s}_t^c / \partial G_t) \geq 1$ (Corollary 4). Once fiscal capacity binds

($\mu_t > 0$, Panel B), the economy becomes vulnerable to amplification: for $\theta_t < 1$ the fiscal-space wedge raises the effective marginal cost of adaptation, so tighter fiscal conditions reduce G_t^* , weaken spread compression and climate-risk mitigation, and further tighten fiscal space (doom-loop region). In contrast, for $\theta_t > 1$, adaptation improves debt dynamics ($\partial B_{t+1}/\partial G_t < 0$), so a binding constraint reinforces rather than discourages investment in G_t (virtuous-cycle region). The narrow band around $\theta_t \approx 1$ highlights that tipping and local multiplicity are more likely when both the fiscal tightness schedule and the policy response are steep (Corollary 6).

This regime map yields three practical levers for managing climate risk under fiscal constraints. First, to avoid the doom-loop region, policymakers can increase the *immediate fiscal payoff* of adaptation by raising ρ_t . In our framework, ρ_t increases when the same unit of adaptation generates larger contemporaneous spread compression, thereby releasing resources absorbed by public and private financing frictions (through Φ and Ψ). Operationally, this points to prioritizing adaptation projects with high and verifiable default-risk mitigation, strengthening implementation credibility (e.g., transparent monitoring, milestone-based disbursement), and designing adaptation programs that are easily priced by markets. These measures raise the marginal spread response ($-\partial s_t^g/\partial G_t$) and ($-\partial \bar{s}_t^c/\partial G_t$), pushing the economy upward in the regime map.

Second, a central objective is to cross the debt-improving threshold $\theta_t > 1$. When $\theta_t > 1$, additional adaptation reduces next-period debt by lowering borrowing costs, so adaptation becomes a *debt-stabilizing* policy rather than a fiscal burden. Accordingly, policy packages that either (i) increase spread sensitivity to adaptation (raising

$-\partial s_t^g / \partial G_t$) or (ii) alleviate rollover pressure and effective debt burden (reducing the likelihood and severity of binding fiscal capacity) help the economy move to the right in Panel B. In this sense, debt-management tools and adaptation policy are complements: improving debt structure and market access creates room for adaptation, while effective adaptation can in turn improve debt sustainability.

Third, in the neighborhood of $\theta_t \approx 1$, managing the risk of tipping requires damping the feedback strength highlighted by Corollary 6. Two mechanisms are particularly relevant: reducing the steepness of fiscal tightness around the constraint (flattening the mapping from debt pressure to μ_t) and reducing the sensitivity of policy to short-run fiscal stress (making G_t^* less elastic to μ_t). Institutional devices that shield critical adaptation spending from procyclical cuts, and contingency financing arrangements that prevent abrupt tightening of fiscal space after shocks, are natural implementations of these mechanisms.

Corporate channel and public-private complementarity. Importantly, the corporate channel enlarges the virtuous region through two distinct effects, as visualized by the upward-rightward shift in Figure 1. By lowering corporate spreads, adaptation releases private resources (raising ρ_t via Ψ), and it improves aggregate financial conditions that feed back into sovereign spreads, increasing θ_t . This public-private complementarity implies that policies enhancing private-sector resilience and access to finance can materially increase the fiscal return to adaptation and help the economy escape the doom loop.

2.6.5. *Why the corporate sector channel enlarges the virtuous region*

The corporate channel strengthens the case for virtuous cycles in two ways. First, it directly enlarges the self-financing region through the additional term $\Psi_{\bar{s}^c}(-\partial \bar{s}_t^c / \partial G_t)$ in (36) as shown in Corollary 4. Intuitively, adaptation that compresses corporate spreads relaxes private financing frictions, freeing up resources that can partly offset (or even dominate) the direct resource cost of G_t .

Second, it shifts the best-response G_t^* upward through the financing-benefit term $\gamma_c K_t(-\partial \bar{s}_t^c / \partial G_t)$ in (34), reducing the sensitivity of optimal adaptation to a binding fiscal-capacity wedge and mitigating the doom-loop mechanism. More specifically, the corporate financing benefit shifts the best-response G_t^* upward for a given fiscal tightness μ_t , as shown in (34). As a result, corporate-spread sensitivity to adaptation (a larger $-\bar{s}_G^c$) expands the set of states in which the economy transitions from the doom-loop region ($B_t(-s_G^g) < 1$) to the virtuous-loop region ($B_t(-s_G^g) > 1$), and mitigates the multiplicity region by weakening the dependence of G_t^* on a binding fiscal constraint.

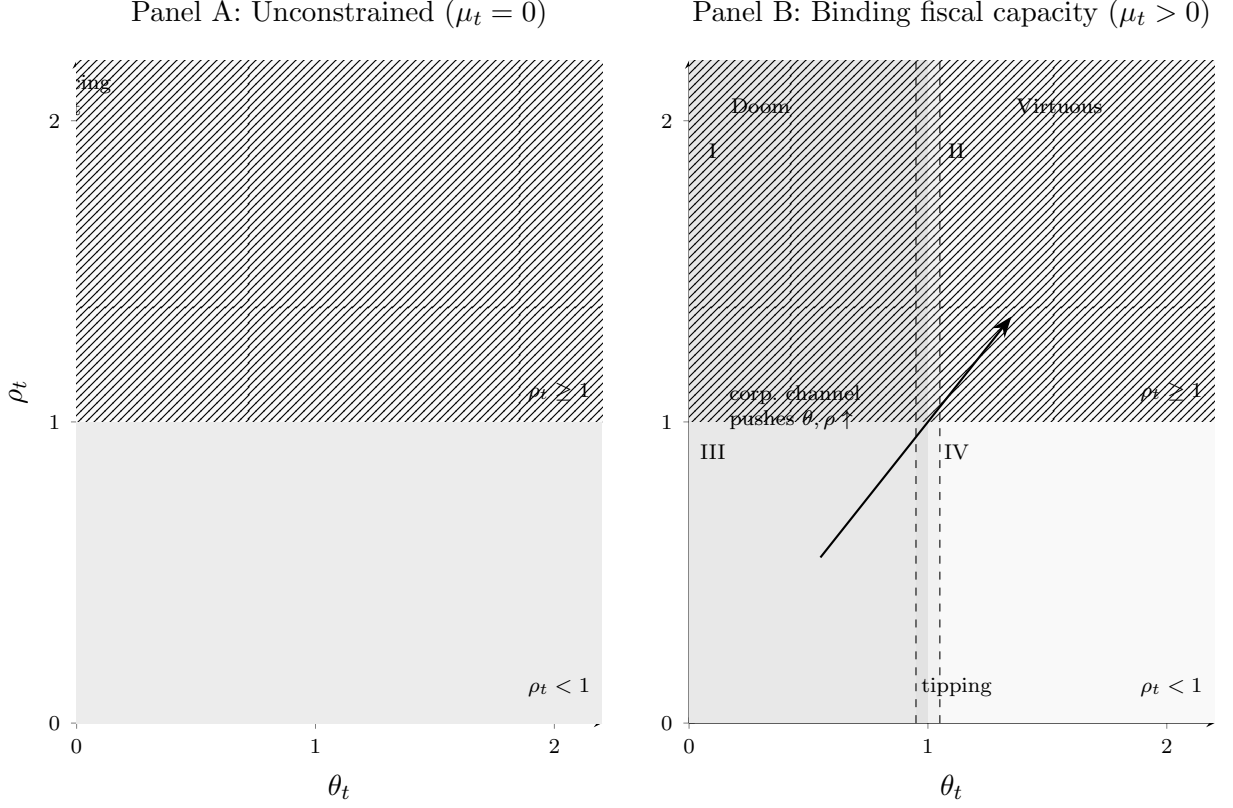


Figure 1: Two-panel regime map accompanying Proposition 7. Define $\theta_t \equiv B_t(-\partial s_t^g / \partial G_t)$ (debt-improving / constraint-relaxing index) and $\rho_t \equiv \Phi_{sg}(-\partial s_t^g / \partial G_t) + \Psi_{\bar{s}^c}(-\partial \bar{s}_t^c / \partial G_t)$ (self-financing-on-impact index). Panel A ($\mu_t = 0$) emphasizes the self-financing threshold $\rho_t = 1$. Panel B ($\mu_t > 0$) highlights the doom vs. virtuous boundary at $\theta_t = 1$ (Proposition 7); the shaded band around $\theta_t \approx 1$ is illustrative of where tipping/multiplicity is more likely under strong feedback (Corollary 6). In Panel B, quadrants I–IV follow the standard convention (upper-left, upper-right, lower-left, lower-right).

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